Python Automata Library (PAL) Cheat Sheet

1-page API map · X... includes Y/Z variants

CONVENTIONS

dims=(nx[,ny[,nz]]) · negative dimension = wrap
1D/2D/3D; omit unused axes. Negative sizes enable periodic boundaries.

hood = array of offset tuples
Neighborhoods contain relative (dx[,dy[,dz]]) offsets.

x... / X... / ItoX... / DispWrapX...
X shorthand includes corresponding Y/Z variants.

i = linear site index · SQ = lattice site · a = agent ID
Coordinates are integer sites or continuous positions.

grid[i] / grid[x,y,z] · slices copy
Index by site or coordinates; slicing returns a copy.

Add(...) then Update() · Reset()
Queued changes apply synchronously.

SafeMode default · pal.FastMode() before construction
Compile kernels with @pal.njit(cache=True).

AGENT GRID · NewAgentGrid(dims,props=0,stack=False)

a=agents.NewAgentSQ(x,y,z) · NewAgent(x,y,z)
Create at lattice site / continuous point.

agents.MoveSQ(a,x,y,z) · Move(a,x,y,z)
Move by lattice / continuous position.

agents.I(a) · XSQ...(a) · X...(a)
Site index, lattice and real coordinates.

agents.Alive(a) · Dispose(a) · GetPop()
Life status, removal, count.

agents[a,p] · agents.counts
Agent properties; read-only occupancy.

agents.AgentsAt(x,y,z) · LastAgent(x,y,z)
Site occupants / last occupant.

agents.AgentsInRadius(rad,x,y,z,exclude=None)
Nearby agents; periodic-aware.

agents.All(shuffle=False)
Snapshot of living agent IDs.

agents.InWrapSQX...(x) · DispWrapX...(p1,p2)
Wrap lattice coordinate; shortest displacement.

LISTS · NewIList() / SAMPLER · NewMultinomial()

lst.Append(i) · Clear() · Random() · Shuffle()
Integer-list operations.

lst.All() · Iter() · len(lst) · lst[i]
Copy / no-copy traversal, indexing.

m.Binomial(n,p) · Setup(n) · Sample(p)
Binomial sampling.

OPENGL WINDOW · StartOpenGLWindow(nx,ny,nz=None,...)

draw.Circle(rad,color,x,y,z=0) · Box(xLen,color,x,y,z=0,...)
Add circles/spheres and boxes.

draw.BoxSQ(color,x,y,z=0) · Line(width,color,x1,y1,x2,y2,...)
Site boxes and segments.

draw.Clear() · win.Clear() · win.Background(color)
Reset primitives / scene / background.

win.Borders(width,color) · win.Camera(x,y,z,yaw=None,pitch=None)
Domain border and 3D camera.

SHARED WINDOW API

win.IsOpen() · Update() · Save(path,block=False) · Close()
Window lifecycle / image output.

win.StartGif(path,delay=100) · AddGifFrame(block=False) · StopGif()
GIF recording; OpenGL supports timeout options.

SHARED GRID API · NewGrid(dims,dtype)

dims=(nx,ny,nz); negative size = wrap
1–3 dimensions; unused axes omitted.

grid.xDim... · nDims · wrapX...
Axis sizes, dimensionality, periodicity.

len(grid) · grid[i] · grid[x,y,z] · grid[...] = value
Site count; read/write; slices copy.

grid.ToI(x,y,z) · ItoX...(i)
Coordinate ↔ linear index.

grid.InWrapX...(x)
Wrap coordinate on periodic axis.

grid.Box(x1,x2,y1,y2,z1,z2)
Half-open rectangular iterator.

grid.Hood(hood,x,y,z,unroll=False)
Offset neighborhood at a site.

PDE GRID · NewPDEgrid(dims)

field.Add(v,x,y,z) · Update() · Reset()
Queue field deltas, apply, clear.

field.SetTimeSpaceStep(dt,dx,dy=1,dz=1) · Dt() · Dx...()
Numerical spacing.

field.Diffusion(rate,BC...) · DiffusionMask(rate,mask,...)
Uniform / masked diffusion.

field.DiffusionField(rates,BC...) · DiffusionInterfaces(rx,ry,rz,BC...)
Cell / face coefficients.

field.DiffusionADI(rate,BC...) · DiffusionRadialCircle/Sphere(rate,outerBC)
Implicit / radial diffusion.

field.Advection(vx,vy,vz,BC...) · AdvectionField(vx,vy,vz,BC...)
Uniform / spatial velocity.

field.AdvectionInterfaces(vx,vy,vz,BC...)
Face-centered velocity.

BC... = xMinBC,xMaxBC,yMinBC,yMaxBC,zMinBC,zMaxBC
Optional face boundary conditions.

POP GRID · NewPopGrid(dims,capacity=None)

pop.Add(v,x,y,z) · Update() · Reset()
Queue count deltas, apply, clear.

pop.GetPop() · All()
Total population; occupied indices.

PIX WINDOW · StartPixWindow(nx,ny,scale=1,...)

pix[x,y]=pal.RGB(r,g,b) · pix[i]=color
Draw packed RGB pixels.

pix.xDim · yDim · ToI(x,y) · ItoX/Y(i) · len(pix)
Buffer geometry.

START / UTILITIES

pal.FastMode() · pal.njit(cache=True)
Select fast mode before object creation; compile model kernels.

pal.Seed(s) · Random() · RandInt(n)
Shared PAL RNG; seed for reproducibility.

pal.MooreHood(dim,excludeCenter=False) · VonNeumannHood(...) · CircleHood(dim,rad,...)
Relative neighbor offsets.

pal.RGB(r,g,b) · GetR/G/B(c) · ColorScale(colors,value)
Pack, extract, interpolate colors.

pal.AwaitWindows(timeout=30)
Wait for rendering windows.